

Network Monitoring (English Notes)

1. What is Network Monitoring?

Imagine your school has a security guard and CCTV operator.

He always watches:

- Who is coming
- Who is leaving
- Where problems are happening
- Who is running around

In networking, this watching process is called Network Monitoring.

It means:

Continuously observing the network

So problems can be detected before they become serious.

Real Example

Suppose the office internet becomes slow.

The network admin checks:

- Who is using too much internet?
- Which device is causing problems?
- Is the router down?

This is called network monitoring.

2. Types of Monitoring

Passive Monitoring

Imagine a teacher only watches the classroom.

The teacher doesn't disturb anyone.

Observe only

It only observes network traffic.

Active Monitoring

Imagine a teacher asks:

"Is everything okay?"

In networking:

The system sends test packets.

Ping → Reply?

It checks whether the network is alive.

Real-Time Monitoring

Like a live CCTV camera.

Seeing what is happening right now.

Examples:

- Router CPU
- Bandwidth
- Online devices

This is called live monitoring.

Historical Monitoring

Checking old records.

Example:

“Yesterday at 2 PM the internet became slow.”

Used for history and analysis.

3. Important Metrics

Bandwidth

Think of it as a road size.

A wider road means more cars can travel.

In networking:

How much data the network can carry.

Latency

Imagine a message takes time to reach someone.

You → Friend

Delay in communication = Latency

Packet Loss

Imagine sending 10 parcels and losing 2.

If data gets lost:

- Calls disconnect
- Games lag
- Videos buffer

Throughput

The actual speed you get.

Example:

Internet speed = 100 Mbps

Actual speed = 70 Mbps

This real speed is called Throughput.

Uptime / Downtime

Uptime = Service is running

Downtime = Service is stopped

4. Monitoring Tools

Wireshark

A detective software.

It captures and analyzes packets.

Examples:

- Who sent data?
- Which IP?
- TCP or UDP?

PRTG

Monitors all devices.

Shows:

- Bandwidth
- CPU usage
- Online/offline status

Nagios

Like a security guard alarm system.

Example alerts:

- Server down!
- Router problem!

SolarWinds

Used for:

- Network monitoring
- Device mapping
- Visualization

Netstat

Shows active network connections on a PC.

Command:

`netstat -an`

5. Real Practice

Check active connections:

`netstat -an`

Check latency using Ping:

`ping google.com`

Capture packets with Wireshark:

- DNS
- TCP
- HTTPS

Key Terms

Monitoring = Watching the network

Bandwidth = How big the road is

Latency = Delay

Packet Loss = Lost data

Uptime = Running

Remember:

Network Monitoring = The CCTV system of a network.

It continuously watches whether the network is working properly or not.